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ECPE 136
19 March 2026

Lab Assignment 5 - 3-input NAND Gate and 3-input NOR Gate

Objective:

- The objective of this lab is to get familiar with using the new software Cadence Virtuoso. Using what we know about designing pmos and nmos schematic, we will design them using the software to then simulate and verify with a waveform.

Procedures:

1.0 - Create a Schematic

- Using Cadence Virtuoso we will create a schematic using pmos, nmos, input and output pins.

2.0 - Create a Testbench Schematic

- Within this test bench schematic we will use the symbol generator to apply our sources to the schematic which includes: vdd, gnd, inputs and outputs. With each input we will apply a voltage pulse to allow for multiple tests when generating a waveform. As well as setting each input with a period delay to have multiple variations when testing for proper output. The vdd and gnd will be constant as vdd is 5v and gnd is 0v. We then select all our inputs and outputs to be measured.

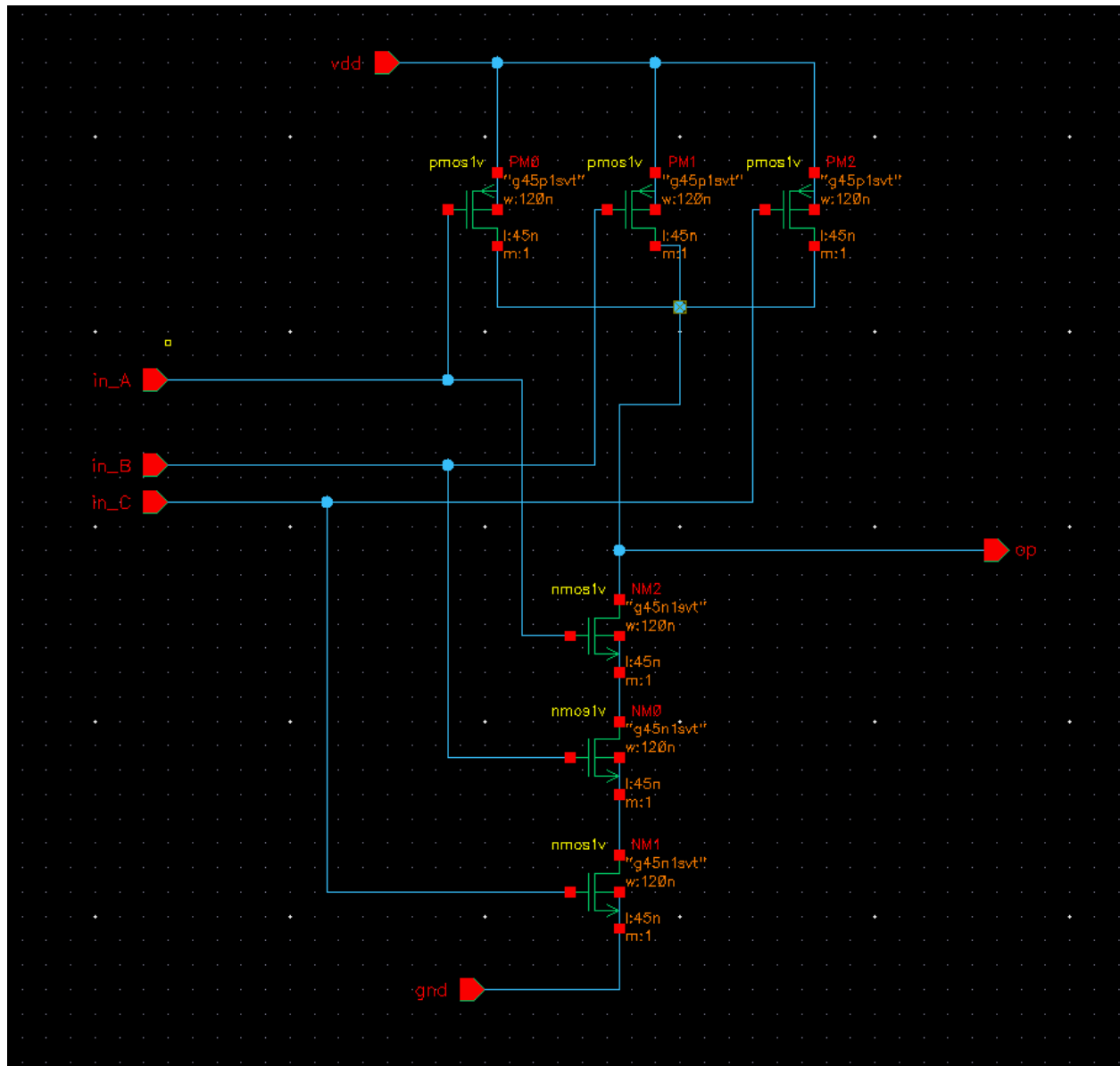
3.0 - Analyze Waveform for Correctness

- Once a waveform has been generated. To verify a correct waveform the number of waves should correspond equally to the number of inputs and outputs. Since in both of these there are 3 inputs and 1 output, there should be a total of 4 waves to compare. Each input can be seen and the output can be analyzed.

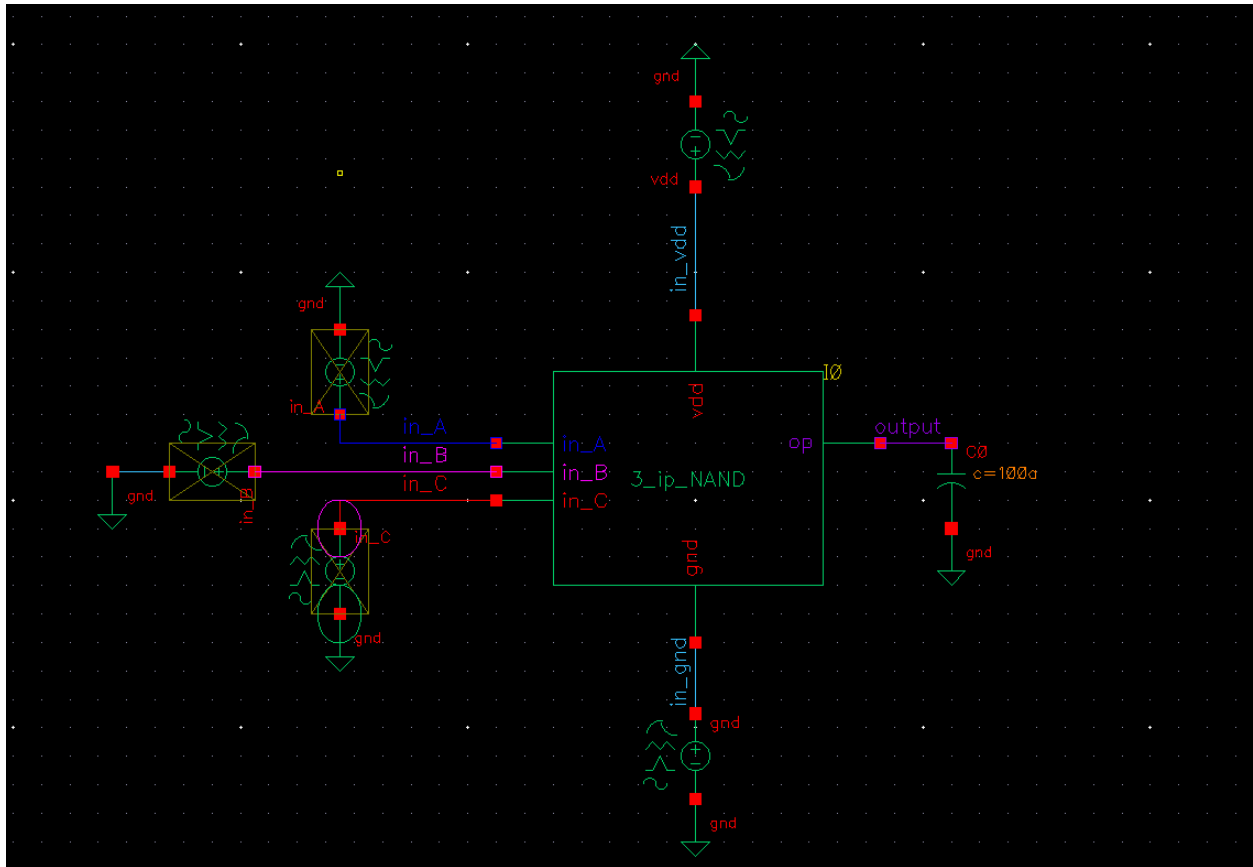
Design description:

Total NAND Transistors: 3 PMOS + 3 NMOS = 6 Transistors

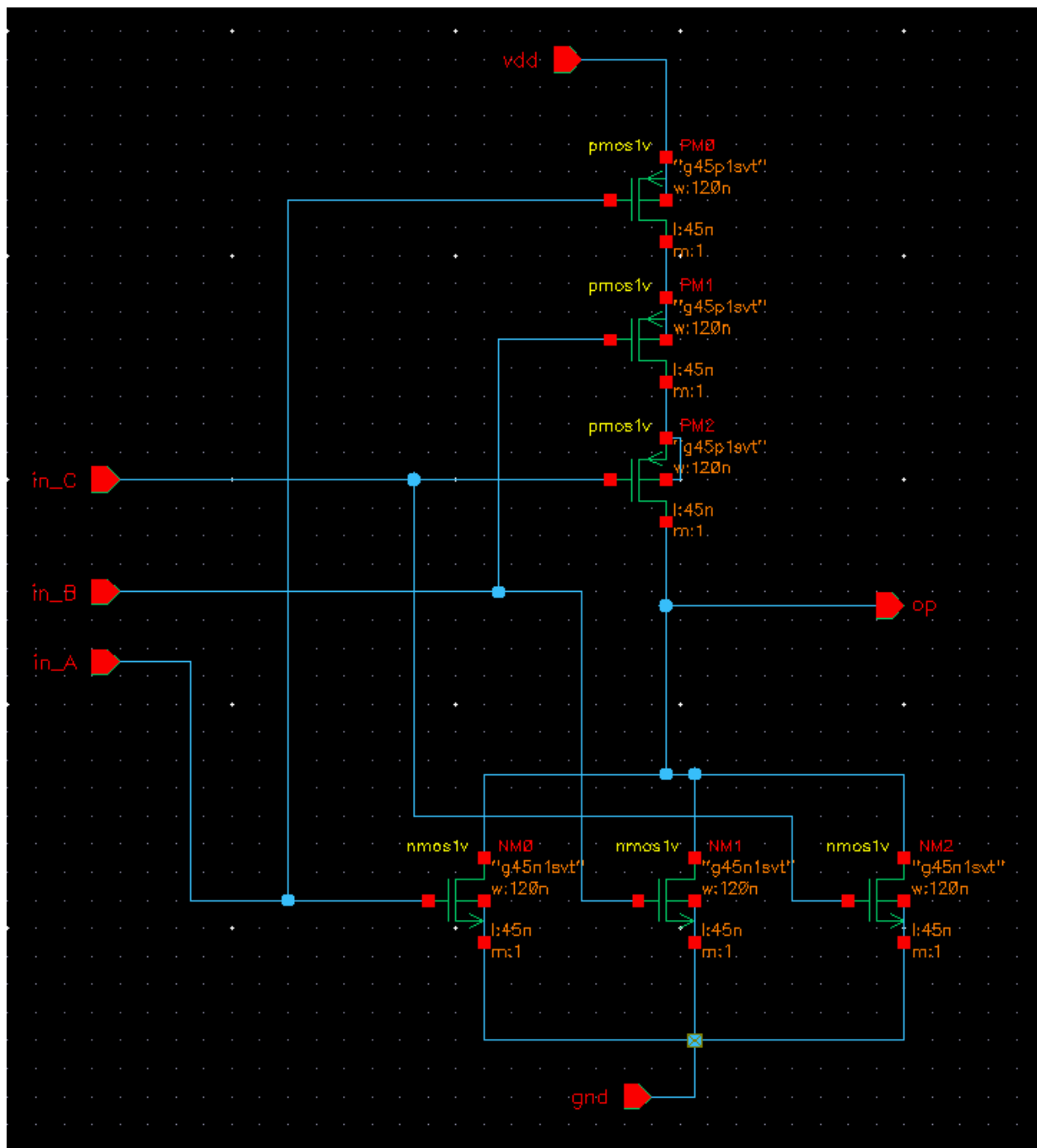
Total NOR Transistors: 3 PMOS + 3 NMOS = 6 Transistors



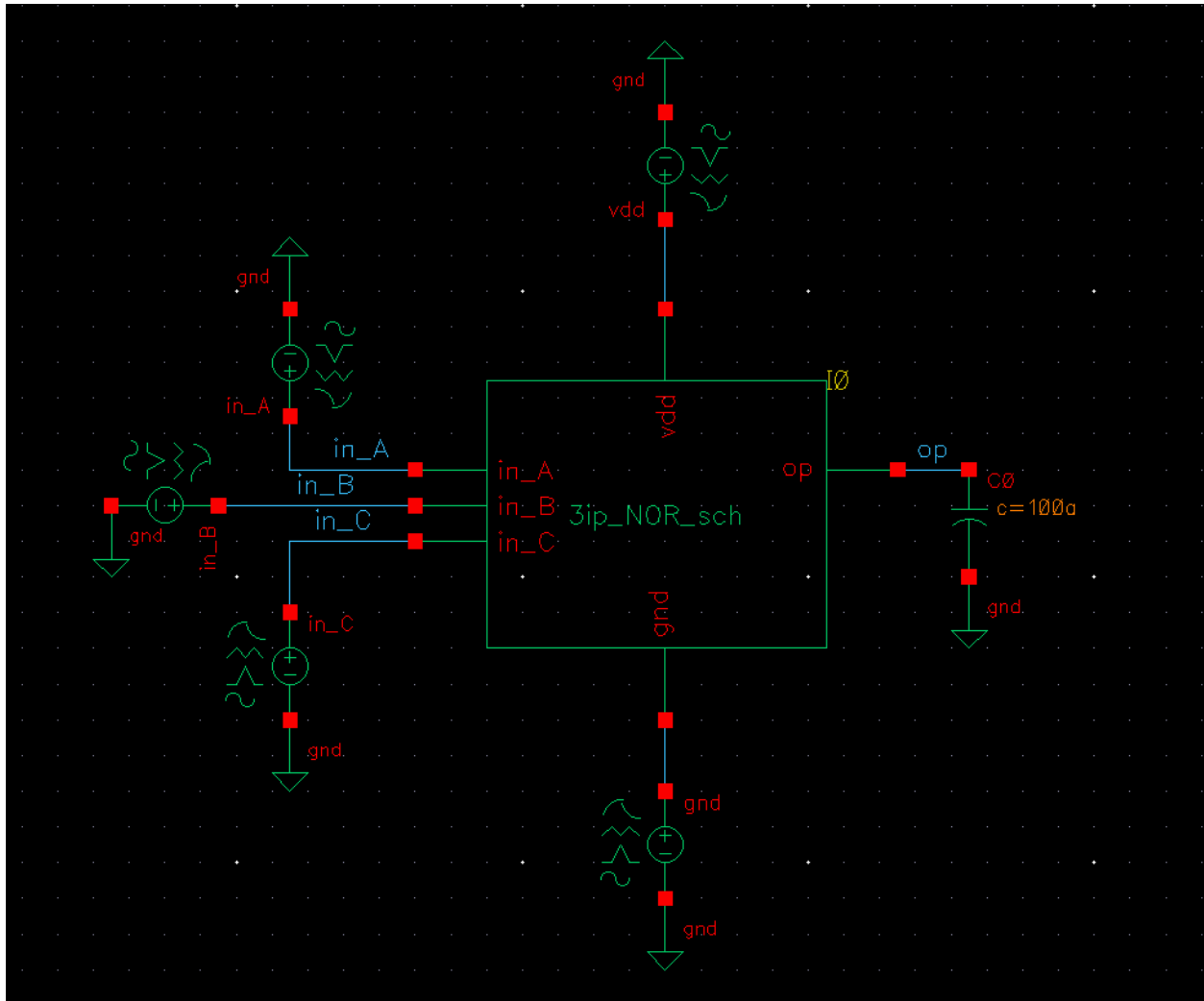
3-Input NAND Schematic



3-Input NAND Test Circuit / Symbol

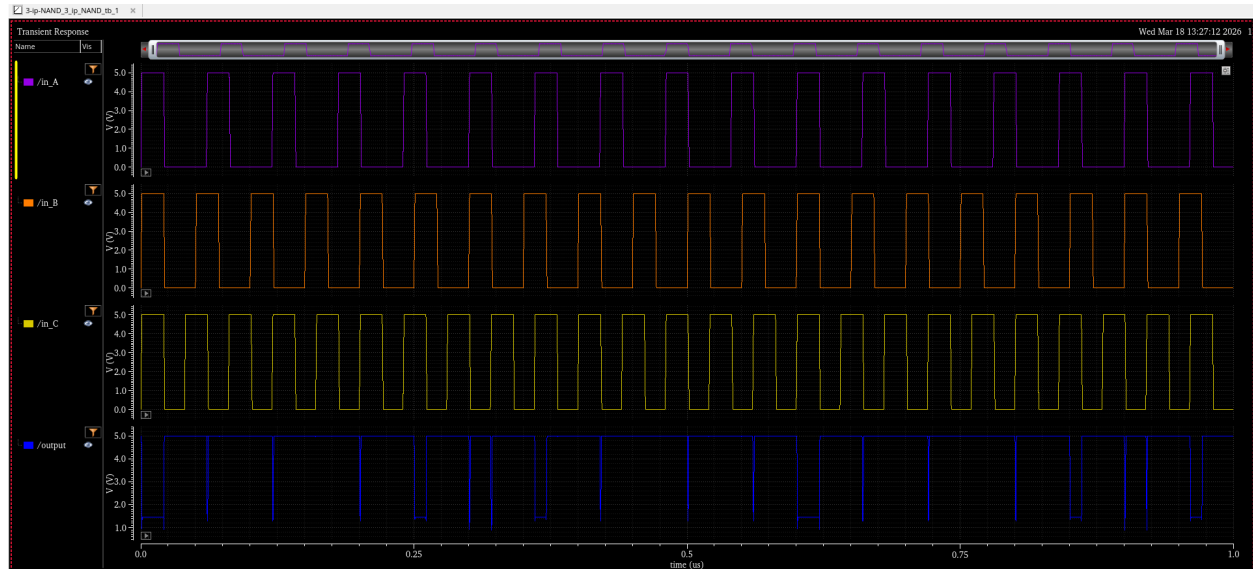


3-Input NOR Schematic



3-Input NOR Test Circuit / Symbol

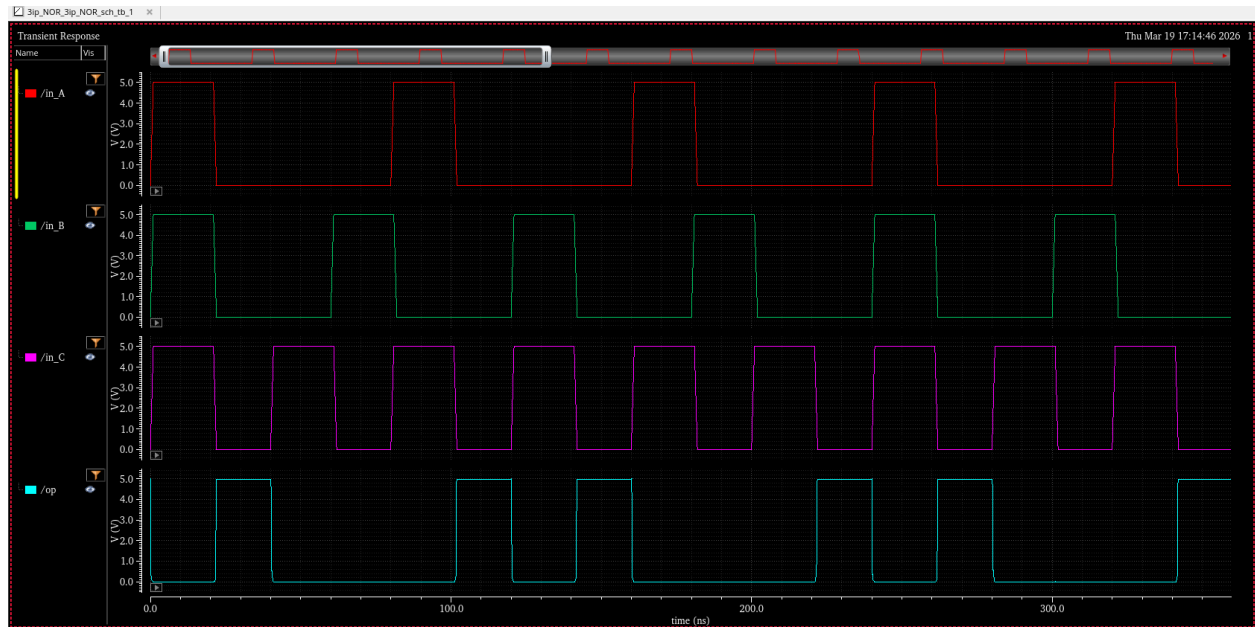
Results & Observations:



3-Input NAND Waveform

Observation:

- When setting up my inputs for the 3-Input-NAND testbench I set each input to have a different pulse width so that each input could result in different variations for testing different inputs. When looking at the graph there is some information we can gather knowing our pmos and nmos. When the pulse is high, this is 1 and nmos is active. When pulse is low 0 and pmos is active. Using this information we can see if the output waveform is correct based on the inputs high or low state. It is accurate to see that the waveform does accurately show correct output. When all inputs are high, the nmos in series are on and a path from ground to output is shown with output waveform being low toward 0v. When looking at any pmos high then a path to vdd and output is made and the output waveform shows a high toward 5v like vdd. Period of Waveform: A:60ns B:50ns C:40ns



3-Input NOR Waveform

Observation:

- Like the 3 input NAND, when setting up values for the inputs, the pulse width differs for each to allow for multiple variations in input values for verification. When considering a NOR circuit like this one. As long as 1 nmos is high the output will point to gnd since they are all in parallel. Since all pmos are in series, all 3 need to be on to point vdd to output. When looking at the waveform we can see that the common output is low or 0, this is because the nmos are in parallel and only need 1 to be turned on to create a connection to output. Period of Waveform: A:80ns B:60ns C:40ns

Conclusion:

- This lab assignment was important because we learned new software and how to use it. Previous assignments used AMD Vivado and taught us to code our logic and verify with a waveform. This time we actually design our schematics and testbenches with the pmos and nmos graphics and connecting wires. This new software allows us to learn how to now visually design and make a schematic and simulate it with various outputs the software will do for us. I find this to be a good tool to use known real pmos and nmos values to make and simulate a circuit. The waveform shows the outputs and how voltage plays a role. Overall learning this software is a next step to becoming better hardware engineers.